

Web Threat Analyzer - Project Report

Abstract

Web Threat Analyzer is a cybersecurity web application that analyzes website security using URL validation, SSL certificate verification, security header analysis, WHOIS lookup, threat score calculation, and MongoDB storage.

Technologies Used

Frontend: HTML, CSS, JavaScript

Backend: Python Flask

Database: MongoDB Atlas

Features

- URL Validation
- SSL Analysis
- Security Headers Check
- WHOIS Lookup
- Threat Score Calculation
- MongoDB Storage

How to Start the Project

1. Open VS Code Terminal
2. Navigate to backend folder:
`cd backend`
3. Start Flask Server:
`python app.py`
4. Open frontend using Live Server.
5. Enter a URL and click Scan Website.

Useful Commands

Install Dependencies:

```
python -m pip install flask flask-cors validators requests python-whois pymongo
```

Run Backend:

```
python app.py
```

MongoDB Connection Test:

Check records in MongoDB Atlas -> Browse Collections -> web_threat_analyzer -> scan_results

Conclusion

The project successfully combines Web Development, Cybersecurity, and Database Management to provide website security analysis and threat assessment.